Algorithms Design and Analysis - HW V

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Due to: 18/01/2026

Please submit the assignment in pairs, and make sure to include both of your names and ids on the PDF. 5 bonus points are given for a typed out solution, and my respect to whoever writes in LATEX. Make sure to prove each claim you use, unless it was seen in class. You may contact me for every question you have about the H.W.

Problem 1

In class, we discussed the complexity of the $Ford-Fulkerson$ algorithm under the assumption that the capacities on edges are integers. In general, we want to analyze its complexity for networks with rational capacities as well. Show how to do this.

Problem 2:

Given a d-regular bipartite graph $G = (V = (A, B), E)$ in which $|A| = |B|$.
Design an algorithm to color the edges of the graph using d colors such that no two adjacent edges (i.e., edges sharing the same endpoint) have the same color.

Problem 3:

Given a flow network $N = (G, s, t, c)$, design an algorithm that finds all the edges that belong to every minimum $s - t$ cut.

Problem 4:

You are tasked with organizing a charity event for $3n$ volunteers. To ensure effective teamwork, the volunteers need to be divided into n teams, each consisting of exactly 3 members.
Among the volunteers, n have been pre-selected as team leaders based on their

experience.
Your goal is to assign each volunteer to a team in such a way that:

1. Every team has exactly one team leader and two additional volunteers.
2. Each volunteer feels comfortable working with their team leader. This is determined by a given friendship matrix, where:
   (a) $Friends[i, j] = True$ means that the i-th team leader and the j-th volunteer are friends.
   (b) $Friends[i, j] = False$ means there is no friendship connection between them.

The algorithm must check if it is possible to assign every volunteer to a team led by a team leader who is their friend. If it is possible, the algorithm should provide a clear assignment of volunteers to their respective team leaders.

Problem 5:

In this problem, we are interested in turning a colorful picture into black and white, assuming the following model:

1. The picture has n pixels.
2. Each pixel v has values w_v, and b_v, indicating the profit we get from setting the pixel white/black.
3. For each pair of pixels u, v, there is a loss function $L(u, v)$, indicating the loss we get by setting u, v to be of different colors.

Under this model, we want to partition the set of pixels V, into two sets, W and B, such that:

$$\phi(W, B) = \sum_{u \in W} w_u + \sum_{v \in B} b_v - \sum_{u \in W} \sum_{v \in B} L(u, v) \tag{1}$$

is maximized. That is, the sum of profits by choosing the color, minus the sum of losses for choosing two pixels to be of a different color, is maximized. You can assume $w_v, b_v, L(u, v) \geq 0 \ \forall u, v$.

Provide an algorithm that partitions V into $W \cup B = V$, $W \cap B = \emptyset$, with the target function maximized. Justify correctness and prove running time.